B300 NICo Bring-up

digi-p2 site · HGX B300 S989423X6713063 · NICo v2.3.0 **Live** · synced 3:06:40 PM

CURRENT STAGE

Run 1 PASSED (Ubuntu 24.04.3 via NICo, NIC mode). Run 2 = NIC mode again with dynamic BMC IP from NICo DHCP; DPU-mode run paused.

Next: Run 2 Phase 0: commit patches, pin unique image tags, back up DB + Vault. Then reset the B300, add a NICo DHCP range (10.10.30.100–.109) and switch the BMC to DHCP.

94% complete Done 16 In progress 0 Blocked 0 To do 0 Skipped 1

Preparation 5/5

1	DONE	**Collect B300 details** BMC MAC/IP, serial, X710 PXE NIC, UEFI mode confirmed; wipe approved. updated 2026-11-03
2	DONE	**NICo control-plane health** 4/4 nodes Ready, NICo v2.3.0, all pods Running, MetalLB VIPs assigned. updated 2026-11-03
3a	DONE	**Network & DHCP design check** vlan30-hostinband (reserved-only) + borderleaf relay .250/.251 already in place. updated 2026-11-03
3b	DONE	**Admin CLI access** nico-admin-cli on tf1 with Vault admin cert (issuer site-root, valid to 3 Dec 2026). updated 2026-11-03
4	~~SKIPPED~~	**DHCP relay on digi-srvr** Not needed: switch relay already forwards to nico-dhcp. updated 2026-11-03

Registration 1/1

5	DONE	**Expected machine for the B300** BMC .77 + X710 .81 reserved, dpu_policy ignore, DPF off, lockdown disabled. updated 2026-11-03

Host takeover 8/8

6a	DONE	**Site Explorer discovers the BMC** Machine fm100psn5d1d7n4... created (Host, Predicted). updated 2026-11-03

6b–6e	DONE	**Fix BMC Redfish errors** BMC firmware 01.07 → 01.12.02.03; Redfish 20/20 OK, 500s gone. updated 2026-11-03
6f-1	DONE	**Root-cause BIOS setup failure** NICo PATCHes /Systems/1/Bios → 405; firmware only accepts /Bios/SD. updated 2026-11-03
6f-2	DONE	**Stage BIOS values in /Bios/SD + reboot** Applied after reboot: QuietBoot off, TXT on, IPv4 HTTP on, IPv4/IPv6 PXE off. New entry Boot0011 = UEFI HTTP IPv4 X710 (905A08C19262). updated 2026-11-04
6f-3	DONE	**X710 HTTP boot entry first in boot order** Boot.BootOrder rejected (400). Set via Supermicro OEM FixedBootOrder (HTTP 202): UEFI Network X710 905A08C19262 first. Applies at next reset. updated 2026-11-04
6g	DONE	**NICo passes platform config, PXE boots scout** Scout booted and DISCOVERY completed (22:18Z, after scout IB-reset fix). Machine left WaitingForDiscovery at 22:19:06Z. updated 2026-11-04 22:19Z
6g-fix	DONE	**Patch NICo BIOS value mapping (bios-sd3) + Re_tryBoot=Enabled** Platform config cleared with the team's patched images (bios-sd2/sd3) + our BIOS settings: SetBootOrder 15:20Z → CheckBootOrder 16:49Z → Measuring/SPDM 16:51Z → WaitingForDiscovery 16:51:18Z (2026-11-04). updated 2026-11-04 16:51Z
6h	DONE	**Validation / attestation → Ready** Machine READY (2026-11-05). Path: Discovery → UefiSetup → Lockdown → MachineValidation (Completed Success) → BootConfiguring/UnlockHost → READY. updated 2026-11-05

OS install 3/3

7a	DONE	**Create Ubuntu templated iPXE OS** Ubuntu 24.04.3 files served from digi-p2-nico-01 via python http.server (systemd unit os-images-http, port 8090): vmlinuz 15,006,088 B, initrd 74,245,462 B, ISO 3,303,444,480 B — all 200 OK from tf1. Docker port publishing on nico-01 does not work (docker0 down). Next: create the templated iPXE OS. updated 2026-11-05
7b	DONE	**OS definition + instance on the B300** OS def 'ubuntu-24.04.3-autoinstall' (f6fca038-2b71-46c4-a2b4-5fab7ebf7278, OS_TYPE_IPXE, org ncx). --os JSON shape: {"variant":{"OperatingSystemId":"<uuid>"},"phone_home_enabled":false,"run_provisioning_instructions_on_every_boot":false,"user_data":"..."}. Allocate needed global flag --cloud-unsafe-op (nico-rest + site agents deployed). Instance 554ad82e-3345-43b7-a24a-adc39ec96bd3 'b300-01_0', flat VPC 7b357b5b, IP 10.10.30.81. updated 2026-11-05

7c	DONE	**PXE install completes; SSH into new OS** Host rebooted → UEFI HTTP → iPXE → vmlinuz/initrd/ISO from 10.11.1.99:8090 → subiquity autoinstall → reboot. Ubuntu 24.04.3 LTS, kernel 6.8.0-147, hostname b300-01, / 3.44 TB, NIC ens3923f0np0 = 10.10.30.81. ssh digi@10.10.30.81 with key OK (2026-11-05 00:04 UTC). `updated 2026-11-05`

Open items & risks

NICo force-restarts the host on every failed BIOS setup
Log: 'machine_setup failed ... triggering force-restart' 10× in 2 h. This is why the node keeps going off/rebooting; not caused by the BMC changes.

Patched nico-api image now bios-sd3 (helm rev 4 kept it)
Helm rev 4 (2026-11-04 17:20) deployed Unbound without reverting nico-api: image is v2.0.0-nico-bios-sd3, pod not restarted. Confirm the tag is in the Helm values so future upgrades keep it.

Lockdown: PATCH HostInterfaces/1 returns 400 on this BMC
NICo 'Setting lockdown and issuing reboot' ×15; PATCH /redfish/v1/Managers/1/HostInterfaces/1 → HTTP 400 ×12. State moved on to MachineValidation anyway; check SysLockdownEnabled afterwards. Machine record kept disable_lockdown=false from creation.

DMI chassis serial is a placeholder (0123456789)
Scout reports dmi chassis_serial '0123456789' (product_serial S989423X6713063 is correct). Fine for now; may matter for SKU/serial matching later. Supermicro can program the FRU.

Chassis $expand takes ~40 s
Works on FW 01.12.02.03; mention to Supermicro.

BMC LAN in Failover, active on Shared port
BMC shares the host's onboard X710 port 0 (the PXE NIC) via NC-SI; dedicated port not linked. Works with the shared VLAN 30 design, but BMC can drop briefly during host reboots/NIC resets (Redfish timeouts). Not the cause of Blocker 1. Better: cable the dedicated BMC port (Failover then switches automatically).

Rotate BMC password
It was shared in chat earlier.

Renew CLI admin cert before 3 Dec 2026
Rerun ~/nico-lab/issue_admin_cert.sh on tf1.

Admin-CLI instance allocate requires --cloud-unsafe-op
Site is wired to nico-rest (rest-api, keycloak, cloud-worker, site-agent x3). Instance 554ad82e was created via admin-cli with --cloud-unsafe-op vinod (labels cloud-unsafe-op, admin-cli-last-modified-by), so nico-rest does not own it. Release it the same way. Production path: allocate through nico-rest API.

Key facts

Machine ID (permanent)	`fm100htcm1d9bklehq3ranjmlr6r6pk4pn1rb5p5jck8ipg65hna7j8hhm0 (was fm100psn5d1d7… predicted)`
BMC	`7C:C2:55:92:68:E7 · 10.10.30.77 · FW 01.12.02.03`

PXE NIC (X710 p0)	90:5A:08:C1:92:62 · 10.10.30.81
Segment	vlan30-hostinband 10.10.30.0/24, gw .254, reserved-only
NICo VIPs	DHCP .105 · PXE .106 · API .116 · REST .120 (10.11.1.x)
Kubernetes	tf1 (CP) + tf2–tf4 · v1.30.4 · installed from digi-p2-nico-01:/root/infra-controller
Platform	Supermicro PIO-422GS-NB3RT · S/N S989423X6713063
Running nico-api image	nvmetal-carbide:v2.0.0-nico-bios-sd3 (pod nico-api-5845679cb6, helm rev 4)
NICo install host	digi-p2-nico-01 (10.11.1.99): /root/infra-controller, KUBECONFIG=~/.kube/tf-poc.config, helm release nico rev 3
Boot DNS	Unbound 10.11.1.108 (UDP+TCP 53): carbide-pxe.forge → 10.11.1.106, carbide-api.forge → 10.11.1.116, forwards to 8.8.8.8
Scout image baseline (before fix)	scout.squashfs 1,936,445,440 B, Last-Modified 2026-10-27 20:57:30 GMT, ETag 6ae1103a.3ae17655-736bd000; nico-pxe image nvmetal-carbide:v2.0.0-nico (pod 8 days old)
Scout image (fix deployed)	scout.squashfs 1,936,490,496 B, Last-Modified 2026-11-04 22:03:04 GMT, ETag 6aebab98.2ae172e9-736c8000; nico-pxe pod 6d6fffd646 (image tag still v2.0.0-nico)
Validation run	ebef13fc-0eff-497d-8d7c-6c4189f96433 · Discovery · 2026-11-04 22:34:50–22:46:21Z · Completed(Success)
OS image URLs	http://10.11.1.99:8090/ubuntu-24.04.3/{vmlinuz,initrd,ubuntu-24.04.3-live-server-amd64.iso}
Tenant / Flat VPC	tenant ncx · bootstrap-flat-vpc 7b357b5b-7fa4-4c55-b1bb-d852660df9c5 (FLAT)

Status is updated by Claude during the session.

Changes by server

tf1 — digi-p2-tf1 · 10.11.1.101

STEP	CHANGE	WHERE	UNDO	BY
3b	Wrapper script to run the CLI inside the pod (no longer	/usr/local/bin/nadm	rm /usr/lo	CLAUDE

	needed)		cal/bin /nadm	
3b	Admin CLI copied out of the nico-api pod (v2.3.0)	/usr/local/bin/nico-admin-cli	rm /usr/local/bin/nico-admin-cli	CLAUDE
3b	Script that issues the admin certificate	~/nico-lab/issue_admin_cert.sh	rm it	CLAUDE
3b	Issued one admin client certificate from Vault (CN nico-api.nico-system.svc.cluster.local, 30 days, until 3 Dec 2026). Vault config unchanged	Vault PKI nicoca/issue/nico-cluster	Let it expire, or revoke by serial in Vault	CLAUDE
3b	Certificate files (dir 700, files 600)	~/.nico/ (admin-cert.json, client.crt, client.key, ca.crt, issue.err)	rm -rf ~/.nico	CLAUDE
3b	CLI config	~/.config/nico_api_cli.json	rm it	CLAUDE
3b	Hosts entry so the API name resolves	/etc/hosts: 10.11.1.116 nico-api.nico-system.svc.cluster.local	Delete that line	CLAUDE
5–6	Saved command output	~/nico-lab/step5_prep.txt, step6_state.txt	rm	CLAUDE
–	No Kubernetes objects, ConfigMaps, Helm releases or pods modified (read-only kubectl only)	Kubernetes	–	NO CHANGE

NICo (via CLI from tf1) nico-api · 10.11.1.116

STEP	CHANGE	WHERE	UNDO	BY
6b	Expected machine: Disable Lockdown false → true	`em patch --bmc-mac-address 7C:C2:55:92:68:E7 --disable-lockdown true`	Same command with false	CLAUDE
6b	site-explorer refresh rejected (already in progress)	`site-explorer refresh 10.10.30.77`	–	NO CHANGE
6e	Remediation resume rejected (machine exists)	`site-explorer remediation --resume`	–	NO CHANGE
6e	Requested a fresh BMC exploration	`site-explorer re-explore 10.10.30.77`	Nothing to undo	CLAUDE
pre-existing	Segment vlan30-hostinband, expected machine (BMC .77, X710 .81, dpu_policy ignore, DPF off), switch relay .250/.251 — set up by the NICo installer, not changed by us	site config / NICo DB / borderleaf1-2	–	NO CHANGE
6g	nico-api image changed outside Helm: v2.0.0-nico → v2.0.0-nico-bios-sd (13:56Z) → v2.0.0-nico-bios-sd2 (14:58Z). Built and pushed to ECR from another session	Deployment nico-api (ReplicaSets 6f66c98d57, 6bd96f8bbc)	`kubectl rollout undo deploy/nico-api -n nico-system` (back to 64845fb69)	SOMEONE ELSE
6g	Unbound enabled via helm upgrade (rev 4): nico-unbound Deployment, unbound-external-udp/tcp LoadBalancer on 10.11.1.108	`helm release nico rev 4 (2026-11-04 17:20)`	`helm rollback nico 3 -n nico-system` (check the	YOUR TEAM

			image tag first)	
6g-fix	nico-api image updated to v2.0.0-nico-bios-sd3 (contents to be confirmed)	Deployment nico-api (pod 5845679cb6)	kubectl rollout undo deploy/nico-api -n nico-system	YOUR TEAM
6h-fix	New scout.squashfs with IB-reset skip fix served by nico-pxe (pod restarted 2026-11-04 ~22:05Z). Image tag unchanged (v2.0.0-nico)	Deployment nico-pxe (pod 6d6fffd646), /public/blobs/internal/x86_64/scout.squashfs	kubectl rollout undo deploy/nico-pxe -n nico-system	YOUR TEAM
6i	Machine fm100htcm1d9bk... reached READY	NICo state machine	–	NICO

digi-p2-nico-01 10.11.1.99

STEP	CHANGE	WHERE	UNDO	BY
–	None confirmed. Only change would be ~/.kube/config if you copied admin.conf there	~/.kube/config (if copied)	rm ~/.kube/config	NO CHANGE
7a	Downloaded Ubuntu 24.04.3 live-server ISO (SHA256 OK) for the NICo OS install	/srv/os-images/ubuntu-24.04.3/	rm -rf /srv/os-images/ubuntu-24.04.3	YOU
7a	Extracted casper/vmlinuz + initrd; serving /srv/os-images on port 8090 with python3 http.server (transient systemd unit os-images-http). Docker container os-images removed (port publishing broken)	/srv/os-images, systemd unit os-images-http	systemctl stop os-images-http	YOU
R2-0	Run 2 backups: source tarball ~/run2-infra-controller-src-2026-11-05.tgz (3.4 GB, mode 600), ~/run2-images-before.txt, ~/run2-values-live.yaml	/root	rm the files	YOU

R2-0	nico-api chart helper line 45 now allows a per-chart image tag (falls back to global). Pins file ~/nico-values-run2-pins.yaml: nico-api bios-sd3@sha256:485732..., boot-artifacts ibskip-amd64@sha256:9b8a44.... Rendered only; NOT applied to the cluster	/root/infra-controller/helm/charts/nico-api/templates/_helpers.tpl (backup .bak-run2), ~/nico-values-run2-pins.yaml	cp _helpers.tpl .bak-run2 _helpers.tpl	YOU
R2-0	Run 2 backups: Postgres pg_dumpall of nico-pg-cluster leader (nico_system_nico, nico_rest, flow, rms) → ~/run2-nico-pg-2026-11-05.sql.gz (191 KB, complete). Vault Raft snapshot → ~/run2-vault-2026-11-05.snap (90,672 B). Both mode 600; temp snapshot removed from vault-0	/root on nico-01	rm the files (they hold secrets; copy off-box to a restricted location)	YOU

digi-srvr (jump host) 10.10.30.19

STEP	CHANGE	WHERE	UNDO	BY
1	Data collection script and output (broken copy in /home/vinod deleted)	~/nico-lab/step1_collect.sh, step1_output.txt	rm	CLAUDE
6a/6e	BMC diagnostics and post-firmware check scripts	~/nico-lab/bmc_diag.sh, post_fw_check.sh	rm	CLAUDE
6f	BIOS snapshot, wanted values, PATCH body	~/nico-lab/bios_now.json, nico_bios_wanted.json, bios_patch.json, /tmp/bios.json	rm	CLAUDE
6f	(If you ran it) saved BMC credentials +	~/.bmc_b300, ~/.bashrc	rm ~/.bmc_b30	YOU

	.bashrc loader line		0; remove the line from ~/.bashrc

B300 (BMC 10.10.30.77 / host) S989423X6713063

STEP	CHANGE	WHERE	UNDO	BY
1b	BMC cold reset	ipmitool mc reset cold	–	CLAUDE
6e	BMC firmware 01.07.00.02 → 01.12.02.03	BMC	Downgrade not recommended	YOU
6f	AC power cycle of the node	PDU / chassis	–	YOU
6f-2	BIOS staged via /Bios/SD (HTTP 202): QuietBoot true→false, TXTSupport Disabled→Enabled, IPv4HTTPSupport Disabled→Enabled, IPv4PXESupport Enabled→Disabled, IPv6PXESupport Enabled→Disabled	/redfish/v1/Systems/1/Bios/SD	PATCH old values back to /Bios/SD, reboot	CLAUDE
6f-2	Host ForceRestart to apply BIOS settings	ComputerSystem.Reset	–	CLAUDE
–	BMC network (static .77, gw .254), DIGIADMIN account and host OS: unchanged (read-only commands only)	BMC / Ubuntu	–	NO CHANGE
6f-2	Second reboot by you; BIOS values confirmed live. UEFI HTTP IPv4 boot entries created (Boot0011 = X710 p0)	/redfish/v1/Systems/1/Bios, BootOptions	–	YOU
6f-3	Boot order via Supermicro FixedBootOrder (HTTP 202): UEFI	/redfish/v1/	PATCH	CLAUDE

	Network (X710 HTTP IPv4 905A08C19262) moved first. Boot.BootOrder PATCH was rejected (400, no change)	Systems/1/Oem/Supermicro/FixedBootOrder (backup: ~/nico-lab/fbo_before.json on digi-srvr)	FixedBootOrder from fbo_before.json, reboot	
6g	Set SNMP.CommunityAccessMode=Limited and SNMP.HideCommunityStrings=true (were null). SNMP stays disabled	PATCH /redfish/v1/Managers/1/NetworkProtocol	No clean undo to null; harmless while SNMP is off	CLAUDE
6g	Host ForceRestart after Unbound deployed; host network-booted scout successfully	ComputerSystem.Reset	–	CLAUDE
6h	NICo set the UEFI/BIOS setup password (site UEFI credential, version 0) and power-cycled the host	BIOS setup password via Redfish (UefiSetup state)	Password is in NICo's Vault (UEFI credential); clear via BIOS/Redfish if the	NICO

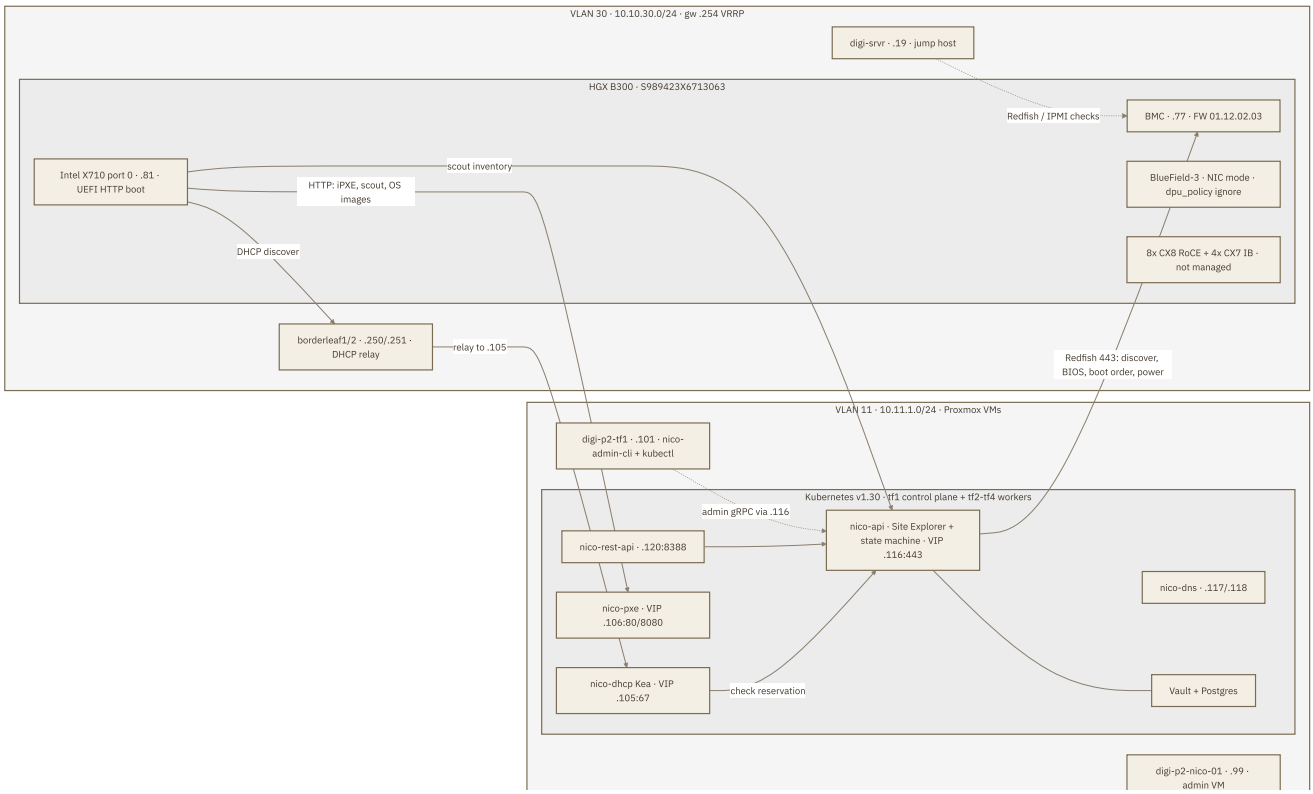
				host leaves NICo

digi-p2-tf1 10.11.1.101

STEP	CHANGE	WHERE	UNDO	BY
7b	Created ~/nico-lab/os/ubuntu-24.04.3.{ipxe,user-data}; nico-admin-cli operating-system create 'ubuntu-24.04.3-autoinstall' (f6fca038…)	NICo DB (operating systems), ~/nico-lab/os	nico-admin-cli operating-system delete f6fca038-2b71-46c4-a2b4-5fab7ebf7278	YOU
7c	nico-admin-cli --cloud-unsafe-op vinod instance allocate → instance 554ad82e-3345-43b7-a24a-adc39ec96bd3 on the B300 (flat VPC, 10.10.30.81); B300 disk wiped and Ubuntu 24.04.3 installed	NICo DB (instances); B300 local disk	nico-admin-cli --cloud-unsafe-op vinod instance release (machine goes through cleanup/wipe back to Ready)	YOU

Technical flow

Topology · who talks to whom, and on which network



Provisioning sequence · from registration to the new OS

